

Tayyab Mahmood Adil

Mechanical Engineering Student | CAD Design & 3D Modeling

03365815973 • t1412530@gmail.com • Islamabad, Pakistan

- [LinkedIn Profile](#)
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ABOUT ME

Mechanical engineering student specializing in **CAD design** and **2D/3D modeling**, with hands-on experience in **SolidWorks** and **AutoCAD**. My goal is to design everything my mind imagines with a proper engineering approach. I am currently **looking for new opportunities** in CAD design and **freelance work**. If you are looking for someone who approaches every project with responsibility and dedication, let's connect!

QUALIFICATION

Air University, Islamabad <i>Bachelor of Mechanical Engineering</i> CGPA: 3.75 (Current)	2025 – Present
Army Public Schools & Colleges, Sarai Alamgir <i>FSC (Pre-Engineering)</i> Result: 87.1%	2023 – 2025
Saint Francis Schools & Colleges, Sarai Alamgir <i>Matriculation (Science)</i> Result: 88%	2021 – 2023

PROJECTS & EXPERIENCE

F1 Helmet Design - Designed using aerodynamic surfacing techniques based on 2D reference blueprints. - Applied complex surface modeling tools to achieve accurate helmet geometry.	SolidWorks 2022
Robotic Leg Mechanism - Modeled a complete robotic leg with full joint assembly and motion simulation. - Simulated range of motion to verify mechanism behavior under movement constraints.	SolidWorks 2022
V-Line 6 Cylinder Engine Model - Recreated internal combustion engine geometry focusing on part tolerances and assembly constraints. - Developed understanding of engine component relationships and manufacturing fit.	AutoCAD 2019
Truss Bridge (3D Printed) - Designed a truss bridge with minimum mass-to-load ratio as the primary optimization target. - Physically 3D printed and tested to validate structural performance.	SolidWorks 2022
Multi-Functional Hammer - Designed and fabricated a hammer, purpose to replace multiple workshop tools. - Gained hands-on experience with workshop machining processes and tolerances.	SolidWorks 2022 + Workshop Project

SKILLS

CAD Software: SolidWorks (Intermediate), AutoCAD 2D/3D (Intermediate), CAD Rendering

Programming: C++ (Intermediate)

Tools & Other: Microsoft Office, Workshop Machining, 3D Printing

LANGUAGES

- **English** (C1- Advanced Academic Proficiency)
- **German** (A-1 In Progress)
- **Urdu** (Native)
- **Punjabi** (Mother Tongue)